

Data Visualization Module (CMP020L013S) - (PG) Spring Term 2022-2023

Coursework

Student Name: Ammar Kafeel
Student ID: KAF-22562988

Task 1: PYTHON VISUALISATION

Visualization 1:

- Question 1:

In what ways do the total cumulative cases and deaths differ between the various WHO regions?

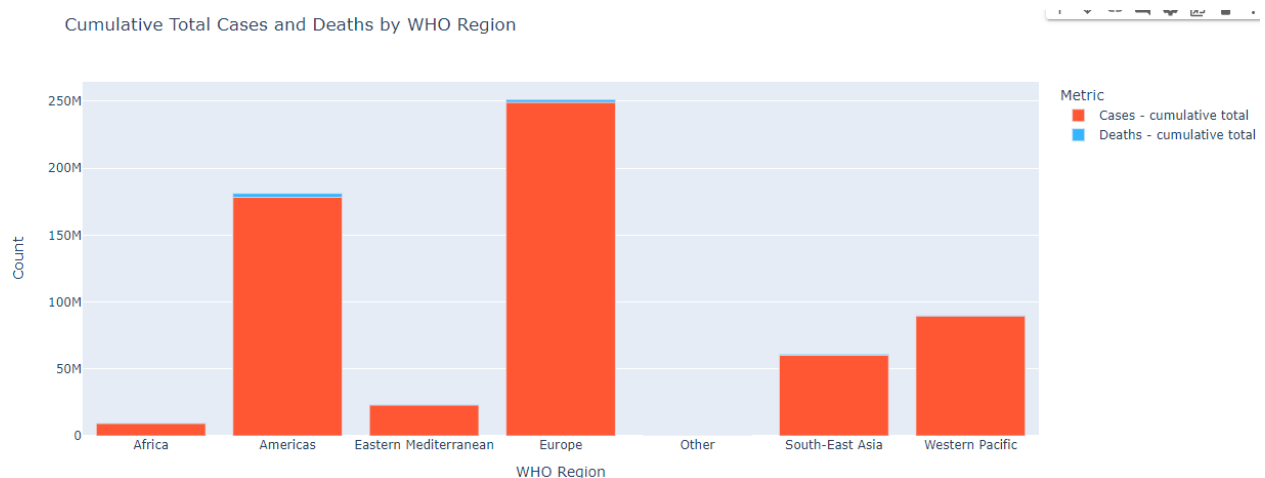
Python code

```
dfw_filtered = dfw.dropna(subset=['WHO Region'])
region_summary = dfw_filtered.groupby('WHO Region')[['Cases - cumulative
total', 'Deaths - cumulative total']].sum().reset_index()

fig = px.bar(
    region_summary,
    x='WHO Region',
    y=['Cases - cumulative total', 'Deaths - cumulative total'],
    title='Cumulative Total Cases and Deaths by WHO Region',
    labels={'value': 'Count', 'variable': 'Metric'},
    color_discrete_sequence=['#FF5733', '#33B5FF'],
)

fig.show()
```

SCREENSHOT



Explanation of the visualization:

An easy way to evaluate cumulative total cases and deaths across WHO areas is made possible with the help of this visualization, it is clear and effective as for the graphic design as well. It is easier to see and distinguish between cases (orange) and deaths (blue) when different colors are used. A simple regional comparison is made possible by the bars on the X-axis that are arranged side by side to reflect the different WHO regions. The Y-axis's bars' lengths graphically represent the extent of the pandemic's effects in terms of cumulative cases and deaths.

I have tried to arrange the plot neatly in terms of visualization, giving a clear summary of the most important details. The placement of the labels and legends is subjected, to reducing visual distractions and improving legibility. Color is used consistently with accepted practices, making the presentation both aesthetically pleasing and easy to understand. I have added the title, axis labels, and color legend to help direct the viewer's understanding even more. All things considered, the plot's deliberate layout and presentation help it effectively communicate important details regarding the cumulative effects of COVID-19 throughout the various WHO areas.

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Visualization 2:

- Question 2:

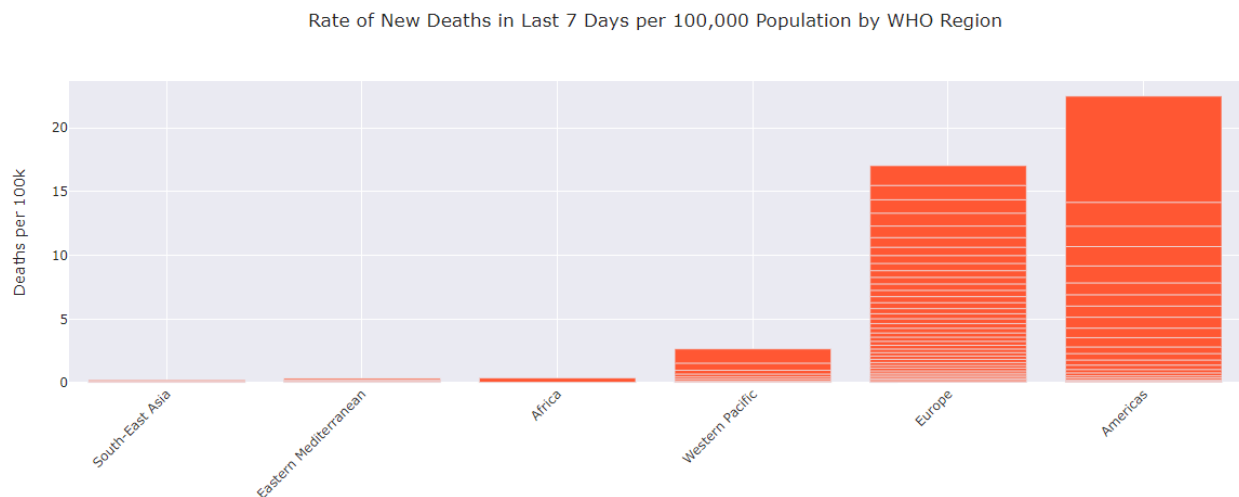
What has been the pattern of newly reported cases per 100,000 people during the past 7 days?

Python code

```
dfw_filtered = dfw.dropna(subset=['Deaths - newly reported in last 7 days', 'Deaths - newly reported in last 7 days per 100000 population', 'WHO Region'])
dfw_filtered = dfw_filtered.sort_values('Deaths - newly reported in last 7 days per 100000 population')
```

```
fig = px.bar(
    dfw_filtered,
    x='WHO Region',
    y='Deaths - newly reported in last 7 days per 100000 population',
    title='Rate of New Deaths in Last 7 Days per 100,000 Population by WHO Region',
    labels={'Deaths - newly reported in last 7 days per 100000 population': 'Deaths per 100k'},
    color_discrete_sequence=['#FF5733'],
)
fig.update_layout(
    xaxis_title='WHO Region',
    yaxis_title='Deaths per 100k',
    xaxis_tickangle=-45,
    showlegend=False,
    template='seaborn',
)
fig.show()
```

SCREENSHOT



Explanation of the visualization:

The rate of new deaths per 100,000 people across the different WHO regions is efficiently communicated by the graphic and this visualization. The various WHO regions are represented by the X-axis, which offers a clear classification for

regional comparisons. Normalized by population, the Y-axis, which highlights "Deaths per 100k," briefly shows how many newly reported deaths occurred in each region during the previous seven days. Viewers may quickly comprehend the design's simplicity and see how different regions have different fatality rates.

The plot clearly shows the structure. Bigger bars indicate areas with relatively greater death rates. Bar heights are used to provide a clear visual representation. the use of coloration is interpreted to highlight the significance of the data itself. This plot focus and ease of digestion are enhanced by the minimalistic approach, which makes it possible for the general public or viewer, to quickly understand the most important information about death rates throughout WHO regions.

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Visualization 3:

- Question 3:

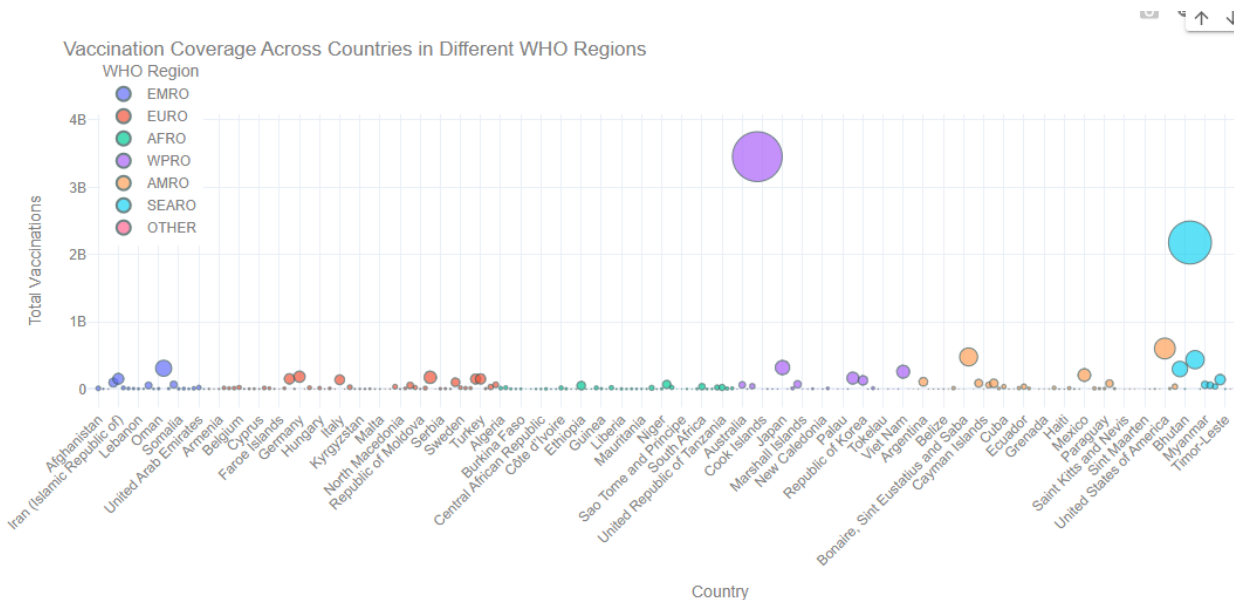
What differences exist in vaccination coverage across various nations and WHO regions (total vaccinations and completely vaccinated)?

Python code

```
dfv_filtered = dfv.dropna(subset=['COUNTRY', 'TOTAL_VACCINATIONS',
'PERSONS_FULLY_VACCINATED', 'WHO_REGION'])
fig = px.scatter(
    dfv_filtered,
    x='COUNTRY',
    y='TOTAL_VACCINATIONS',
    size='PERSONS_FULLY_VACCINATED',
    color='WHO_REGION',
    title='Vaccination Coverage Across Countries in Different WHO
Regions',
    labels={'TOTAL_VACCINATIONS': 'Total Vaccinations',
'PERSONS_FULLY_VACCINATED': 'Fully Vaccinated'},
    size_max=30,
    opacity=0.7,
    template='plotly_white',
```

```
)
fig.update_traces(marker=dict(line=dict(width=1, color='DarkSlateGray')),
selector=dict(mode='markers'))
fig.update_layout(
    xaxis_title='Country',
    yaxis_title='Total Vaccinations',
    xaxis_tickangle=-45,
    showlegend=True,
    legend_title='WHO Region',
    legend=dict(
        yanchor="top",
        y=1.2,
        xanchor="left",
        x=0.01,
    ),
    font=dict(family='Arial', size=12, color='gray'),
)
fig.show()
```

SCREENSHOT



Explanation of the visualization:

A thorough overview of vaccination coverage worldwide is provided by this scatter plot in a way, which also shows the distribution of total vaccines and the proportion of people who have received all recommended immunizations in each

nation. The Y-axis shows the total number of vaccinations given in each country, while the X-axis shows specific countries and provides a thorough analysis of their vaccination programs. A nuanced view of overall vaccination coverage is provided by the size of the marks, which correlates to the number of people who have received all recommended vaccinations. It is made easier to compare vaccination practices and advancements across different geographic areas quickly and informatively by using distinct marker colors that correspond to different WHO regions.

Plotting is done with deliberate attention to detail for the viewer; countries with effective immunization efforts are highlighted by larger markers. Improved marker visibility is a result of both the careful adjusting of marker border opacity and the addition of dark slate gray lines, especially in regions where data points overlap. By effectively classifying WHO areas by color, the legend above the plot reduces interference with the data points. From this visualization, viewers can derive important strategic insights that will help them better evaluate the effectiveness of immunization programs across different regions and pinpoint locations where vaccination rates differ.

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Visualization 4:

- Question 4:

Is there a relationship between the overall number of immunizations administered per 100 individuals and the types of vaccines used?

Python code

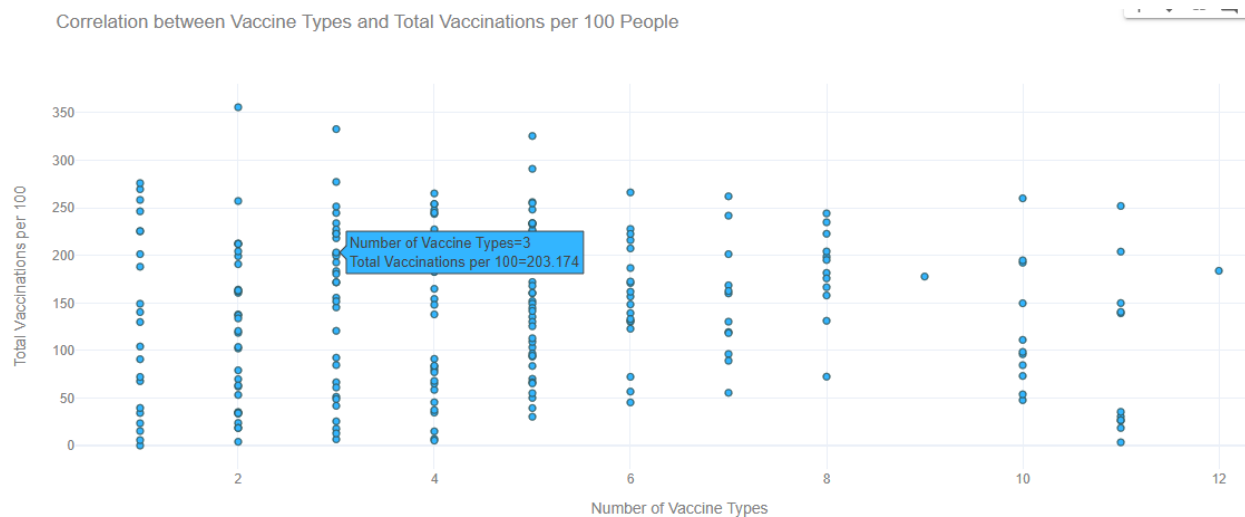
```
dfv_filtered = dfv.dropna(subset=['NUMBER_VACCINES_TYPES_USED',
'TOTAL_VACCINATIONS_PER100'])
fig = px.scatter(
    dfv_filtered,
    x='NUMBER_VACCINES_TYPES_USED',
    y='TOTAL_VACCINATIONS_PER100',
    title='Correlation between Vaccine Types and Total Vaccinations per
100 People',
```

```

labels={'NUMBER_VACCINES_TYPES_USED': 'Number of Vaccine Types',
'TOTAL_VACCINATIONS_PER100': 'Total Vaccinations per 100'},
color_discrete_sequence=['#33B5FF'],
template='plotly_white',
)
fig.update_traces(marker=dict(line=dict(width=1, color='DarkSlateGray')),
selector=dict(mode='markers'))
fig.update_layout(
axis_title='Number of Vaccine Types',
yaxis_title='Total Vaccinations per 100',
showlegend=False,
font=dict(family='Arial', size=12, color='gray'),
)
fig.show()

```

SCREENSHOT



Explanation of the visualization:

A detailed understanding of the correlation between a country's immunization rate per 100 people and the variety of vaccines it uses can be obtained by interpreting the scatter plot. A thorough analysis is made possible by the plot's point distribution, which makes it possible to spot trends, clusters, and patterns. Notably, the nations on the upper right may represent effective vaccination

efforts because they have a better coverage rate per 100 individuals and a bigger number of vaccine kinds. These observations can help draw well-informed conclusions regarding the benefits or disadvantages of using different vaccine types.

This visualization makes it easier to understand how the range of vaccines available and the coverage of vaccination programs by population relate to each other. It also gives health authorities a visual tool to help them make decisions about global immunization strategies.

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Visualization 5:

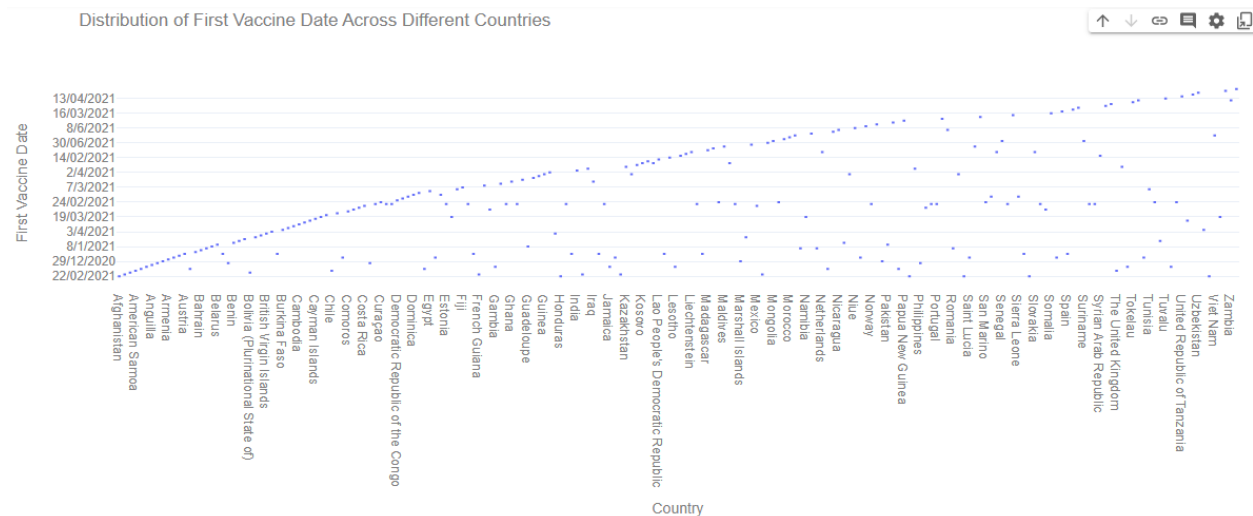
- Question 5:

How is the first vaccination date distributed among the many nations?

Python code

```
dfv_filtered = dfv.dropna(subset=['COUNTRY', 'FIRST_VACCINE_DATE'])
fig = px.box(
    dfv_filtered,
    x='COUNTRY',
    y='FIRST_VACCINE_DATE',
    title='Distribution of First Vaccine Date Across Different Countries',
    labels={'FIRST_VACCINE_DATE': 'First Vaccine Date'},
    template='plotly_white',
)
fig.update_layout(
    xaxis_title='Country',
    yaxis_title='First Vaccine Date',
    font=dict(family='Arial', size=12, color='gray'), # Font settings
)
fig.show()
```

SCREENSHOT



Explanation of the visualization:

The plot is shown as a series of box plots, each of which represents a different country and shows the first vaccination date distribution's interquartile range (IQR). The middle line within each box indicates the median date of the first immunization. Data points that are longer than the whiskers on the box represent outliers, which are countries where vaccination efforts were started much earlier or later than in most other countries.

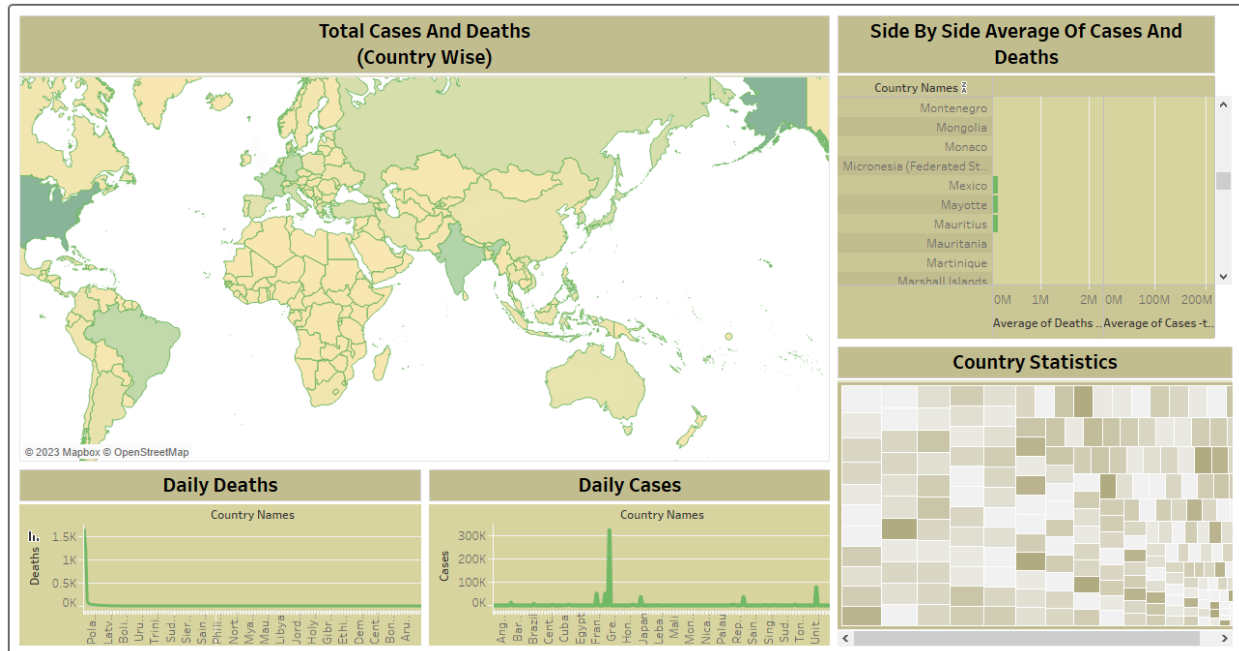
This box plot gives a clear picture of the global distribution of vaccination start dates by demonstrating the temporal variation in the start of vaccination campaigns in various countries. This picture is useful not only for seeing trends and patterns in the early phases of vaccination campaigns but also for assessing the temporal dimensions of the global vaccination

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Task 2: TABLEAU VISUALISATION

Dashboard:

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Introduction:

The goal of the WHO COVID-19 Data Visualization Dashboard is to give consumers a thorough and informed understanding of the ongoing COVID-19 pandemic. Using information directly from the official WHO COVID-19 dashboard, this visualization attempts to provide a concise and current depiction of important parameters including cases, deaths, and immunizations among various nations and areas [1].

Key Insights of Dashboard:

1. Trends of Daily Cases and Deaths: Showcase the daily patterns of COVID-19 cases and fatalities so that crucial times and possible hotspots can be found.

2. Vaccination Heat Map: Examine the range of vaccine kinds that are used in various nations and regions together with the total number of vaccines that are given.
3. Moving Averages of Deaths and Cases: Evaluate the averaged trends in deaths and cases to have a better idea of the whole impact.
4. Maps of Total Cases and Deaths: These provide a spatial overview of all COVID-19 cases and deaths, making it easier to pinpoint areas that will be most affected.

Users can obtain important insights into the dynamics of the COVID-19 pandemic through the dashboard I created, which will help them comprehend the worldwide response to the problem better.

Sheet 1:

Line Chart of Daily Deaths

- **Title:**
Daily Deaths
- **Introduction:**
The daily death tolls for every country are shown in this chart.
- **Crucial Elements:**
Time-series trend line chart.
- **Interactivity:**
Users can observe precise daily death counts by hovering over data points.
- **Key Takeaways:**
 - Global Impact:** 1,668 deaths every day are recorded globally, giving a global picture of the pandemic's daily toll.
 - Major Losses:** The United States is unique in that it has a high daily death toll of 1,220, indicating the severity of the crisis in the nation.
 - Regional Impact:** The daily death tolls in the Russian Federation and Chile are 106 and 49, respectively, suggesting that the impact varies depending on the region.

These findings demonstrate the seriousness of the pandemic's worldwide effects and the difficulties that individual nations have in controlling and reducing COVID-19-related deaths.

Sheet 2:

Line Chart of Daily Cases

- **Title:**
Daily Cases
- **Introduction:**
The daily case counts for each nation are shown in this figure.
- **Crucial Elements:**
Time-series trend line chart.
- **Interactivity:**
Specific daily case counts can be seen by hovering over data points.
- **Key Takeaways:**
Global Spread: 325,602 cases are reported every day worldwide, indicating the pandemic's broad reach.
Impact on the US: With 79,340 cases per day, the US presents a sizable number of cases each day, highlighting the considerable difficulties in managing and suppressing the virus in the nation.

The European landscape continues to be affected by COVID-19, as evidenced by the close follow-up of Germany and France, with daily case counts of 51,364 and 51,348, respectively. These findings provide insight into the regional and worldwide dynamics of the pandemic, directing future research and countermeasures to slow the virus's spread.

Sheet 3:

Heat Map of Vaccination

- **Title:**
Country Statistics
- **Introduction:**

Data about vaccinations is visualized in this heatmap.

- **Crucial elements:**

A heat map with countries displayed, colors denoting the variety of vaccines used, and sizes denoting the overall number of vaccinations.

- **Interactivity:**

Users can investigate the total number of vaccines and vaccine kinds administered in each nation.

- **Data Source:**

Vaccine Records [1].

- **Visual Indications:**

The amount of vaccination types is shown by color intensity. Total immunizations are indicated by the size of each cell.

- **Key Takeaways:**

These insights illustrate the number of vaccine types used and the related overall vaccination rates per 100 people, giving an overview of nations with differing vaccination coverage levels. Notable instances include Cuba and Gibraltar, which have high vaccination rates per 100 inhabitants, while other nations, such as Jordan, have lower rates, indicating a variety of international immunization initiatives.

Sheet 4:

Bar Chart with Moving Average

- **Title:**

Side by Side Average of Cases and Deaths.

- **Introduction:**

The moving average of cases and deaths for every nation is shown in this bar chart.

- **Crucial Elements:**

Comparing moving averages in a bar chart.

- **Interactivity:**

Users can investigate the moving averages over various time frames.

- **Key Takeaways:**

It is simple to see the average trends in cases and deaths over time by looking at the bars. It makes it possible to compare nations quickly and

determine whether one has had notable shifts in their case or death rates. As a smoothing tool, the moving averages provide a more stable picture of the pandemic's overall effects across all nations. This graph helps identify nations with significant changes or new trends and provides insight into the overall trajectory of COVID-19 cases and deaths.

Sheet 5:

Total Cases and Deaths Maps

- **Title:**
Total Cases and Deaths (Country Wise)
- **Introduction:**
The total cases and deaths for each nation are shown on this map.
- **Crucial elements:**
A map showing the total number of cases (colored by size) and deaths (indicated by color).
- **Interactivity:**
For additional in-depth information, users can click on the various countries.
- **Visual Cues:**
Total cases are represented by color intensity. Marker size reflects the total number of deaths.
- **Key Takeaways:**
The map gives a geographical overview of all COVID-19 cases and fatalities worldwide. Higher concentrations of cases or deaths are indicated by denser colors in particular places. Hovering over a country allows users to see comprehensive information, such as the precise number of cases and deaths for each area. This chart provides a visual summary of how the pandemic's effects are distributed across various geographical areas. Areas with high total case counts or noteworthy fatality rates can be rapidly identified thanks to the differing color intensity. Users can focus on particular regions and evaluate the severity of the pandemic's effects in various parts of the world with the use of this graphic, which helps to comprehend the spatial distribution of COVID-19 cases and deaths.

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References:

- [1] (No date) *World Health Organization*. Available at: <https://covid19.who.int/data> (Accessed: 10 December 2023).